Journal of Revenue and Pricing Management (2026) 25:368–377
https://doi.org/10.1057/s41272-026-00574-9

RESEARCH

Potential overinterpretation of results in the abstracts of machine learning studies for movie box office revenue prediction: a systematic review

Joerg Leukel[1] · **Zejun Liu**[2] · **Vijayan Sugumaran**[3]

Received: 12 May 2025 / Accepted: 29 January 2026 / Published online: 16 February 2026

Abstract

Machine learning (ML) has significantly advanced the prediction of movie box office revenue (MBOR). However, ML studies are subject to the risk of overinterpretation, defined as the misuse of language so that evaluation results appear overly positive. This phenomenon is particularly acute in the abstract, which serves as the primary tool for attracting readership, thereby creating the potential for overemphasis on the most favorable results. To assess the frequency and prevalence of overinterpretation in study abstracts, we conducted a prospectively registered systematic review of the MBOR literature. Three databases (Scopus, IEEE Digital Library, and ACM Digital Library) were searched for English-language peer-reviewed articles published until 2024. We evaluated 46 eligible articles for nine reporting practices adapted from a classification of overinterpretation in biomedical literature. The most prevalent practices in the abstract were the omission of performance metrics (87%) and inappropriate use of strong and leading words (43%). The absence of mean absolute percentage error (MAPE, 30%) and coefficient of determination (R^2, 24%) in the abstract was also common, even though these metrics were provided in the main text. Contrary to expectation, no inappropriate extrapolation of evaluation results in abstracts was found. This review highlights the diversity of reporting practices in the abstracts of ML-based MBOR studies. We submit recommendations for enhanced reporting, which can promote the accurate interpretation of evaluation results and the synthesis of evidence from multiple studies.

Keywords Artificial intelligence · Predictive analytics · Regression · Reporting quality · Spin

Introduction

Machine learning (ML) is increasingly being used to predict continuous variables that are characterized by high variability and are subject to a high degree of uncertainty (Gautam et al. 2021; Phumchusri and Phupaichitkun 2024). One such variable is movie box office revenue (MBOR), defined as the total sum of a film's box office takings. Predicting MBOR is important for producers, studios, and investors to make informed decisions about budgeting, marketing strategies, and distribution (Eliashberg et al. 2006). Additionally, theaters and streaming platforms rely on MBOR forecasts to plan screenings, allocate resources, and negotiate licensing deals, which supports effective content delivery to audiences. The extreme uncertainty of MBOR results from viewer behavior in the target markets (Vany 2003). At the same time, a wealth of metadata, such as genres, budgets, professional critics' reviews and consumer engagement behavior on social media, as well as reliable revenue data from industry associations, are available (Oh et al. 2017; Ravid 1999). The challenge lies not so much in collecting data for training prediction models but in constructing and selecting features from the input data to develop accurate and robust models (McKenzie 2023). It is therefore not

✉ Joerg Leukel
joerg.leukel@uni-hohenheim.de

Zejun Liu
rhonda@ctbu.edu.cn

Vijayan Sugumaran
sugumara@oakland.edu

[1] University of Hohenheim, Stuttgart, Germany

[2] Chongqing Technology and Business University, Chongqing, China

[3] Oakland University, Rochester, USA

surprising that some studies provide remarkable evaluation results for ML prediction models. For instance, one study indicated in the abstract that the mean absolute percentage error (MAPE) was only 4.1% using nationwide data from the U.S. (Li and Liu 2022).

Although the results for MBOR prediction appear promising, extensive care in the reporting of results is required so that prospective users of the models can properly assess their validity and generalizability. One potential deficiency in the reporting is the so-called 'overinterpretation' or 'spin', defined as the misuse of language that affects how readers interpret the findings. This misuse may be deliberate or accidental. If the reporting of evaluation results does not "faithfully reflect the nature and range of findings" (Boutron and Ravaud 2018, 2613), their interpretation by readers may be distorted such that the results are portrayed as better than they really are. This phenomenon has been extensively examined in the biomedical literature (Boutron and Ravaud 2018; Yavchitz et al. 2016). Indeed, overinterpretation of findings was identified as a prevalent phenomenon in ML-based clinical prediction models in oncology (Dhiman et al. 2023) as well as in crop production (Leukel et al. 2025). Given its crucial role as the primary tool for attracting readership and peer audience (Cals and Kotz 2013), the abstract is particularly acute for this phenomenon, thus creating the potential for overemphasis on the most favorable evaluation results.

Based on the biomedical literature and a previous review (Leukel et al. 2025), four groups of misleading reporting practices for assessment within the abstract can be identified. First, *inconsistency in reporting of evaluation results between abstract and main text* can mislead the reader by hiding important performance metrics that are only available in the main text (Dhiman et al. 2023). Specifically, an abstract might claim high levels of performance without providing any performance metric to support this claim. Second, *unclear interpretation of performance metrics* draws conclusions based on unreliable statistical analysis (Andaur Navarro et al. 2023; Kempf et al. 2018), such as using correlation coefficients instead of performance metrics for regression tasks. Although this deficiency is assessed in the main text, it represents a foundational error that underlies and potentially validates subsequent problematic, overly positive reporting in the abstract. Third, *inappropriate use of strong and leading words* can mislead the reader's interpretation through "using overly optimistic and positive words" to describe the model's performance (Andaur Navarro et al. 2023, 102). Qualitative claims made with words such as 'accurate', 'excellent', 'robust', etc. must be supported by the reporting of corresponding qualitative performance metrics. Fourth, *inappropriate extrapolation of evaluation results* is a form of misleading extrapolation, as the abstract might claim transferability of the findings to wider or different settings (Andaur Navarro et al. 2023; Kempf et al. 2018), such as application to other movie-related variables.

While previous reviews of ML studies for MBOR prediction provide important insights into the processes for data collection, preprocessing, training, and evaluation (Ahmad et al. 2020c; Mbunge et al. 2022), the reporting quality, particularly the potential for overinterpretation of evaluation results within the abstract, has not been examined so far. It is essential to know the prevalence and manifestations of overinterpretation in this area, so that such studies are interpreted with appropriate caution. This knowledge can also raise awareness of the dangers of overinterpretation and help to ensure that future abstracts report their results more completely, accurately, and correctly. Against this backdrop, the present review aims to assess the frequency and prevalence of overinterpretation in the abstracts of articles reporting evaluation results for MBOR prediction models. Specifically, we adapted a classification of overinterpretation approach proposed in the biomedical literature and applied it to the MBOR prediction literature by conducting a systematic review.

Method

The present review was registered in advance. All procedures were carried out as defined in the registration, which can be accessed at https://osf.io/f3d5n. There were three minor deviations from the registered protocol; these are clearly described in the following subsections. The review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2021) (Page et al. 2021) as appropriate, with details provided in the PRISMA checklist (Supplementary Material 1). We provide the collected data at Figshare(https://figshare.com/s/84e2b0ced06b4efb9a71 https://doi.org/10.6084/m9.figshare.30931817); details on the dataset structure are provided in Supplementary Material 4d.

Eligibility criteria

Studies were eligible if they evaluated at least one ML-based model for predicting the box office revenue of movies, without restriction to any film types, countries or time periods. We defined the following eligibility criteria: (1) prediction model for MBOR (e.g., quarterly revenue in US dollars); (2) prediction model used a supervised learning algorithm (e.g., Artificial Neural Networks); (3) prediction model was evaluated using real data (e.g., from Box Office

Mojo by IMDbPro); (4) English-language article published until 2024.

Studies were excluded if the target variable was categorical, such as classification of movie success (e.g., blockbuster, hit, and flop), or other continuous variables, such as movie ratings in review portals. In addition, mechanistic or deterministic prediction models—that is, models which did not learn relationships between input data and the measured outcome (MBOR) but instead relied on predefined mathematical functions—were not eligible. Furthermore, we excluded articles that did not report a quantitative evaluation or used synthetic (simulated) data. Moreover, articles published after 2024 and non-English articles were not considered.

Information sources and search strategy

Records were retrieved from three electronic databases on December 15, 2024, namely, Scopus, ACM Digital Library and IEEE Digital Library. We selected these databases for their comprehensive coverage of journal articles and computer science proceedings. Scopus, in particular, was chosen because it has a larger coverage of peer-reviewed literature than the Web of Science (Singh et al. 2021; Thelwall and Sud 2022). This selection enables high replicability by utilizing databases with structured indexing. We deliberately excluded Google Scholar due to its lack of structured export and transparent search mechanisms, and DBLP due to its limitations in metadata detail and scope outside of computer science. The search was performed on the title, abstract, and keywords fields. The search query included various terms for MBOR (9 terms), prediction (4 terms), and machine learning (24 terms) as well as limiters for publication year, language, and document type (where applicable). For each database, we adjusted the query to the database syntax. The full queries for each database are provided in Supplementary Material 2. Table 1 shows the elements of the query for Scopus.

Table 1 Search query for scopus (limiters not shown)

Element	Search term
Fields	TITLE-ABS-KEY
Component 1: Target variable	(“movie revenue*” OR “movie gross revenue*” OR “movie sales” OR “cinema box office revenue*” OR “film revenue*” OR “box office revenue*” OR “box office earnings” OR “box office sales” OR “box office outcome*”)
Component 2: Prediction	(assess* OR estimat* OR forecast* OR predict*)
Component 3: Machine learning	(“artificial intelligence” OR bagging OR convolutional OR “decision tree” OR “deep learning” OR “elastic net” OR “ensemble learn*” OR “fuzzy logic” OR “Gaussian process” OR “gradient boost*” OR “kernel regression” OR KNN OR LASSO OR “linear regression” OR “machine learning” OR “nearest neighbo*” OR “net regression” OR “PLS regression” OR PLSR OR “partial least” OR “random forest*” OR “ridge regression” OR “support vector” OR “neural network*”)

Selection process

The records retrieved from each database were imported and merged into a spreadsheet. One investigator identified duplicate records based on the Digital Object Identifier (DOI) field, and in cases of missing DOI, checked the article title field. For the exclusion of articles, we used a coding form and instructions, which were initially tested on a random sample of three records. The coding instructions are provided in Supplementary Material 3. The titles, abstracts, and keywords were independently reviewed by two coders to determine if a predefined exclusion criterion was clearly fulfilled so that the records should be removed. This screening was conducted in rounds of twenty records. The inter-rater agreement for the screening of records, assessed using Cohen’s Kappa, was $\kappa = 0.74$, indicating substantial agreement. Disagreements between coders were resolved through discussion after each round. After this first screening, we downloaded the full texts of the retained articles. The same coders independently screened the full texts to verify compliance with all eligibility criteria (ten articles in each round), and resolved inconsistent codes through discussion after each round. Again, inter-rater agreement was substantial $\kappa = 0.70$.

Data collection process

Two coders independently extracted data from each article using a spreadsheet form (available in Supplementary Material 4a) and coding instructions (Supplementary Material 4b), which were initially developed by the main investigator and adapted from a previous review (Leukel et al. 2025). Coders were instructed to record any qualifying comments in square brackets immediately following the coded field, which were then utilized during conflict resolution. We pilot-tested the form and instructions with two randomly drawn articles (ID 23 and 34). The coders used privately stored Excel sheets to record their codings, which were eventually uploaded to a shared secured platform. The extraction was focused on information necessary to assess the reporting practices within the included articles and it was conducted in rounds of ten articles. The percent agreement for the 11 data fields assessed across the 44 non-pilot studies was 94%. The agreement for individual fields was as follows: 100% for fields 1, 6, 7, and 10; 98% for fields 3, 4, and 11; 93% for field 5 and 8; 89% for field 9; and 68% for field 2. All discrepancies in the extracted data were resolved by the coders through discussion after each round.

One supplemental field related to the evaluation metrics reported in the main text was extracted by the main investigator and recorded separately as 'ReportedMetrics' in the final dataset; this field was not defined in the pre-registered protocol. Three bibliographic data items—specifically, the source title, year of publication, and source type (journal or conference proceedings)—were collected separately from the literature databases during the study selection phase and were not part of the manual coding process.

Data items

Two coders manually extracted data for 11 specific fields per article using the spreadsheet form provided in Supplementary Material 4a. These fields constituted the raw data needed to determine the status of the nine reporting practices central to this study. The detailed operationalization, showing the derivation of the nine reporting practices from the raw data fields, is provided in Supplementary Material 4c. Table 2 presents the reporting practices and states how they can lead to overinterpretation of evaluation results. To define reporting practices, we adapted a classification of overinterpretation proposed in previous research in biomedical literature (Kempf et al. 2018; Lazarus et al. 2015) and organized the reporting practices in four groups. In a previous study, we have already used a similar adaptation for another area of application (Leukel et al. 2025).

Table 2 Definition of reporting practices. (adapted from Leukel et al. 2025)

Group	Reporting practice	Overinterpretation: Article could…
Inconsistency in reporting of evaluation results between abstract and main text	1) Abstract does not report any performance metric.	…hide low levels of predictive performance.
	2) Abstract does not report R^2 but only in the main text.	…hinder readers from comparing the results with results of similar studies.
	3) Abstract does not report MAPE but only in the main text.	…hinder readers from comparing the results with results of similar studies.
Unclear interpretation of performance metrics	4) Article interprets R^2 as linear correlation.	…draw conclusions based on a misinterpretation of R^2.
	5) Article interprets correlation coefficients as a measure of predictive performance.	…draw conclusions based on statistics that do not measure predictive performance.
Inappropriate use of strong and leading words in the title or abstract	6) Title uses such words.	…exaggerate the predictive performance.
	7) Abstract uses such words.	…exaggerate the predictive performance.
Inappropriate extrapolation of evaluation results in the abstract	8) Abstract suggests application to a movie-related variable other than revenue.	…suggest applying the model to a variable for which the model has not been evaluated.
	9) Abstract suggests application to revenue prediction other than for movies.	…suggest applying the model to a variable for which the model has not been evaluated.

MAPE = mean absolute percentage error; R^2 = coefficient of determination

Inconsistency in reporting of evaluation results between abstract and main text is a form of misleading reporting, as it can mislead the reader by hiding important performance metrics that are only available in the main text (Dhiman et al. 2023). We operationalized this using three practices: omission of any performance metric (RP1), inconsistency regarding the coefficient of determination (R^2, RP2), and inconsistency regarding MAPE (RP3). We focused on R^2 and MAPE as these are frequently used, unitless metrics for regression tasks.

Unclear interpretation of performance metrics was assessed in the main text as a form of misleading interpretation, as it draws conclusions based on unreliable statistical analysis (Andaur Navarro et al. 2023; Kempf et al. 2018). Although assessed outside the abstract, this deficiency represents a foundational error that underlies and potentially validates subsequent problematic, overly positive reporting in the abstract. We coded two practices: Incorrect interpretation of R^2 as indicating a linear relationship between observed and predicted revenue (RP4); and mistakenly viewing correlation coefficients as indicators of predictive performance (RP5) (Sheiner and Beal, 1981).

Inappropriate use of strong and leading words is another form of misleading interpretation because it uses "overly optimistic and positive words to describe the model or the model's performance" (Andaur Navarro et al. 2023, p. 102). We assessed titles (RP6) and abstracts (RP7) separately but examined the context in which statements were made. Coders were given the following initial list of words: 'accurate', 'cost effective', 'efficient', 'excellent', 'good', 'novel', 'precise', 'reliable', 'robust', 'significant', and 'superior'. Coders looked for base forms, inflected forms, and derived forms and they could mark additional words that were used inappropriately.

Inappropriate extrapolation of evaluation results is a form of misleading extrapolation, as the abstract might claim that the evaluation results apply to a wider context than the one actually studied (Andaur Navarro et al. 2023; Kempf et al. 2018), such as application to other movie-related variables (RP8) or revenue prediction for non-movies (RP9).

Data synthesis

We used descriptive statistics to summarize the extracted data. Specifically, we determined both the absolute counts and the relative proportions for each reporting practice. We also calculated a cumulative sum score to represent the total number of practices present in each article. Furthermore, deviating from our pre-registered protocol, we conducted a subgroup analysis to investigate potential differences in reporting practices based on the publication venue (journals vs. conference proceedings) and the year of publication, and a co-occurrence analysis of reporting practices. Microsoft Excel 365 was used for data management and IBM SPSS Statistics 29 was used for the statistical tests.

Results

Study selection

Figure 1 presents the process of study selection using a PRISMA diagram. The database search yielded 86 records from Scopus, 24 records from IEEE Digital Library, and 6 records from ACM Digital Library. In the first screening of 95 unique records, we excluded 33 records. The full-text screening assessed 62 articles of which 46 were eligible. We provide the lists of included and excluded articles in Supplementary Material 5 and 6.

Study characteristics

Eight articles were published in 2020, followed by six articles in both 2022 and 2023, and five articles in 2021. The oldest article dates from 2010. Eighteen articles appeared in conference proceedings and 28 articles in journals.

Reporting practices

The frequency and prevalence of the nine reporting practices are shown in Table 3. As a measure of overall prevalence of overinterpretation within individual studies, the total number of distinct reporting practices per study (range 0 to 9) had the following distribution: 2 articles (4%) had 0 practices (Lu et al. 2014; Zaidi and Urolagin 2021), 13 articles (28%) had 1 practice, 19 articles (41%) had 2 practices, 10 articles (22%) had 3 practices, and 2 articles (4%) had 4 practices. The mean number of reporting practices identified per study was 1.93 (SD = 0.93).

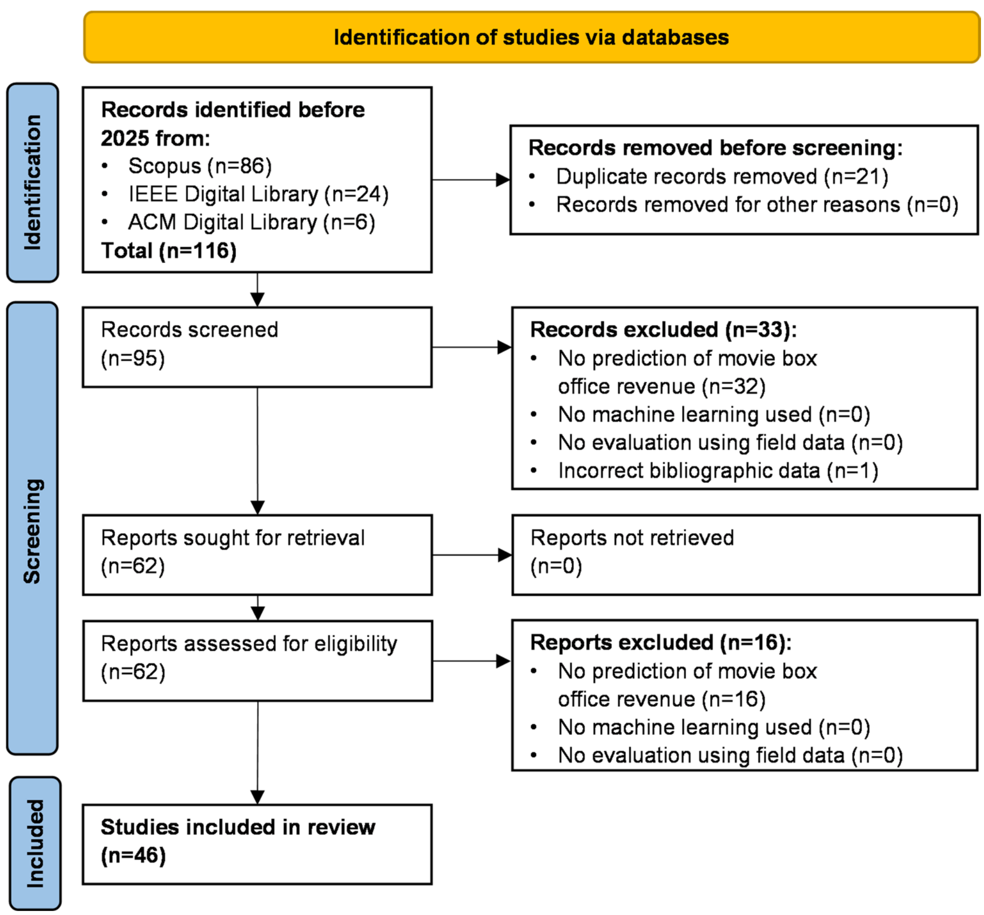

Fig. 1 Study selection process

Table 3 Frequency and prevalence of reporting practices ($n=46$)

Reporting practice	n	%
1) Abstract does not report any performance metric.	40	87
2) Abstract does not report R^2 but only in the main text.	11	24
3) Abstract does not report MAPE but only in the main text.	14	30
4) Article interprets R^2 as linear correlation.	0	0
5) Article interprets correlation coefficients as a measure of predictive performance.	5	11
6) Title inappropriately uses strong and leading words.	0	0
7) Abstract inappropriately uses strong and leading words.	20	43
8) Abstract suggests application to a movie-related variable other than revenue.	0	0
9) Abstract suggests application to revenue prediction other than for movies.	0	0

Inconsistency in reporting of evaluation results between abstract and main text

Although every article's main text reported evaluation results using one or more performance metrics, 40 abstracts (87%) did not report any performance metric. For instance, a study by Bogaert et al. (2021) reported root mean square error (RMSE) and mean absolute error (MAE) as scale-dependent metrics as well as MAPE and R^2 as unitless metrics but omitted them in the abstract. Overall, 15 studies evaluated prediction models using MAPE, but only one of the abstracts reported this metric (Li and Liu 2022), resulting to inconsistent reporting in 14 abstract (30%). Similarly, 12 studies assessed performance using R^2, and only one of the respective abstracts provided this metric (Zaidi and Urolagin 2021), resulting in inconsistent reporting in 11 abstracts (24%).

Unclear interpretation of performance metrics

In cases where R^2 was reported in the abstract or main text, we checked how the authors interpreted this metric. There were no misinterpretations as a linear correlation. Supplementary Material 7a describes how R^2 was defined and used in the relevant studies ($n=12$). In our data extraction, we noted that one article incorrectly specified R^2 as the formula for the Pearson correlation coefficient but with the two components in the numerator squared (Ruan et al. 2015) and another formula had a wrong variable in the numerator (Xiao et al. 2016).

We found that five main texts (11%) incorrectly treated the correlation between observed and predicted revenue as an indicator of performance, whereby larger coefficients were considered to indicate higher predictive performance. Quotes from these texts are provided in Supplementary Material 7b. One of these articles even claimed that the correlation coefficient is one of three metrics "most commonly used in evaluating movie revenue predictions approaches" (Ahmad et al. 2020a, p. 672).

Table 4 Examples of how strong and leading words were used inappropriately in the abstract

Statement (Highlights added by the authors for emphasis)	Reason
"[…] both least squares support vector regression and recursive partitioning strategies **greatly outperform** dimension reduction strategies and traditional econometrics approaches in forecast accuracy, there are further **significant gains** from using hybrid approaches." (Lehrer and Xie 2022, 189)	It is unclear to what extent the strategies outperform other approaches and what the gains are as the abstract does not report any performance metric.
"Our experimental results prove the **superiority** of our approach compared to three baseline approaches and achieved a relative absolute error of 29.65%." (Ahmad et al. 2020b, p. 1)	It is unclear what the baseline approaches are as the abstract does not specify these approaches.
"The precise predictions […] demonstrate the **efficiency and versatility** of our proposed method." (Li and Liu 2022, 1)	Although performance metrics are reported in the abstract, this study examined neither the efficiency nor the versatility of the prediction models.
"[…] evaluation with a Root Mean Squared Error (RMSE) of 14,625.2142, indicating that the model can predict revenue **accurately**." (Madongo et al. 2023, 1406)	It is unclear whether the RMSE is low because the unit of measurement (is it US dollars or Chinese RMB?) and distribution parameters of observed revenue are missing.

Inappropriate use of strong and leading words

We found inappropriate use of strong and leading words in 20 abstracts (43%). In our assessment, we analyzed the context in which the words were used, i.e., whether the results reported in the abstract supported the statement made. We mapped specific forms of words and phrases onto their generic form (e.g., "very good accuracy" and "better than other models" were mapped onto "good"). Among others, we found the following generic forms: significant ($n=3$), accurate ($n=3$), improve ($n=2$), superior ($n=2$), efficient ($n=1$), outperform ($n=1$), and versatile ($n=1$). Table 4 shows four example statements and why they represent a misinterpretation of evaluation results.

Subgroup analysis

Differences in the frequency of the nine reporting practices (RP1 to RP9) between studies published in journals and conference proceedings were assessed using Fisher's exact test (2-sided). There were no statistically significant differences between the two publication venues: RP1 ($p=0.380$), RP2 ($p=0.080$), RP3 ($p=0.522$), RP5 ($p=1.000$), and RP7 ($p=1.000$). Reporting practices 4, 6, 8, and 9 were not included in the analysis as their frequency was zero in both subgroups. Furthermore, the sum score of all reporting practices did not significantly differ between journals ($M=1.82$, $SD=0.95$) and conference proceedings ($M=2.11$, $SD=0.90$), as indicated by the Mann-Whitney U test ($U=212$, $p=0.342$, 2-tailed).

To assess the presence of temporal differences in the prevalence of reporting practices, we first conducted binary logistic regression with publication year as the independent variable. None of the five analyzed RPs showed a statistically significant temporal trend: RP1 ($p=0.942$), RP2 ($p=0.355$), RP3 ($p=0.727$), RP5 ($p=0.149$), and RP7 ($p=0.338$). We then calculated the Spearman rank-order correlation between the publication year and the total sum score per study. This score tended to decrease over the years, however, the association was weak, $r_s(44) = -0.287$, $p=0.053$ (2-tailed). A specific comparison of the sum score between recent years (2022–2024, $M=1.64$, $SD=0.63$, $n=14$) and earlier years (2010–2021, $M=2.06$, $SD=1.01$, $n=32$) showed no statistically significant difference ($U=163.5$, $p=0.127$, 2-tailed). Overall, we found no evidence that the prevalence of reporting practices has declined in recent years.

Co-occurrence analysis

To determine if specific reporting practices tend to appear together non-randomly, we conducted Fisher's exact tests for the co-occurrence of all 10 unique pairs among the five practices with non-zero frequencies (RP1, RP2, RP3, RP5, and RP7). The analysis revealed no statistically significant co-occurrence between any pair, with p-values ranging from 0.120 to 1.000.

Discussion

Summary of main results

We analyzed forty-six articles on ML models for MBOR prediction for reporting practices in the abstract that indicate whether evaluation results are overinterpreted. We found omission of performance metrics in the abstract (87%), inappropriate use of strong and leading words in the abstract (43%), inconsistency between abstract and main sections regarding MAPE (30%) and R^2 (24%), whereas inappropriate extrapolation of evaluation results was not observed.

Implications

The results of this review are important for how individual MBOR prediction studies are interpreted. Our findings—especially the high prevalence of omissions and strong words—necessitate that readers from academia and industry, including film studios, distributors, and financial analysts, approach these abstracts with heightened caution. The findings are worrisome as the abstract should help readers quickly grasp the significance of the study results and decide on continuing to read on or not (Shiely et al. 2024). The abstract's crucial role as the primary, and often only, point of access means that it must present standardized metrics that can be easily interpreted by researchers and practitioners alike. Given the significant financial stakes in MBOR prediction, readers must be aware that positive linguistic framing may lack quantitative support and must actively seek out the complete performance metrics in the main text to assess the commercial utility and risk of a model. This caution ensures that models are not over-relied upon for high-stakes investment decisions.

The widespread incomplete reporting in the abstract presents a challenge for the building of knowledge through incremental research and future meta-analyses on MBOR prediction models. The MBOR field is particularly well-suited for systematic synthesis, given the relative availability of similar data across studies, including objective revenue data from industry associations as the predicted variable and viewer ratings as raw data for feature definition. Despite this unique commonality in training and evaluation data, the high rates of omission and inconsistency make direct comparison and synthesis between studies nearly impossible based on abstracts alone. This poor reporting creates an untapped potential for meta-analysis, which could otherwise inform more focused improvements in model development and provide benchmarking for the community.

Authors should be more aware of how natural language and quantitative results, i.e., performance metrics, must be used consistently and faithfully to avoid overinterpretation. Many words that are frequently used for describing the quality of a prediction model – such as accurate, efficient, robust, and significant – refer to clearly defined, quantitative measures; hence, these words must be used in this sense, and any use of comparative or superlative must be supported by empirical evidence. If there is any uncertainty in this regard, it is better to omit the linguistic qualification and instead leave the interpretation of the performance metric to the reader. Unitless metrics are more suitable for comparing models from different studies than metrics with units such as the MAE and RMSE. The latter metrics can only be understood if the statistical parameters of MBOR are also known (Chicco et al. 2021). For example, a given

MAE value can indicate a low or high error, depending on measures of centralization and dispersion of the target variable. Therefore, we suggest to report the mean and standard deviation of MBOR. If authors want to avoid this effort, a more efficient solution would be to report only the unitless metrics, since they already take into account the dispersion of the underlying variable and perform a standardization (or normalization) in a meaningful way. In addition to MAPE and R^2, the so-called normalized RMSE (NRMSE) can also be used here, in which the RMSE is divided by the standard deviation. This metric is referred to as RMSE-observations standard deviation ratio (RSR) in other literature (Moriasi et al. 2007). However, no study in our review reported this metric.

Reviewers and editorial boards have a crucial role in mitigating overinterpretation in abstracts at an early stage. Reviewers should pay close attention to ensuring that evaluation results are presented completely, accurately and correctly in articles, and that this is done consistently between abstracts and main sections. To achieve this, it may be necessary to establish review teams possess both high expertise in ML and statistics and a heightened awareness of misleading language. Furthermore, editorial boards could greatly reduce the risk of omission by strictly requiring abstracts that mandate the reporting of key performance metrics. Its implementation should not be hindered by restrictive limitations on the length of abstracts, especially since the space required for extended reporting is relatively small.

Recommendations and guidelines for reporting on ML studies have been proposed in various fields, such as biology (Walsh et al. 2021), clinical research (Vasey et al. 2022), environmental research (Zhu et al. 2023), and information systems (Kühl et al. 2021). Adopting such guidelines can at least help to increase the completeness of how evaluation results are reported. The development of such guidelines is most advanced in the biomedical literature, applying to various study types and offering comprehensive detail. For example, the TRIPOD+AI for Abstracts checklist defines 13 items, including item 11 requiring authors to "report model performance estimates (with confidence intervals)" (Collins et al. 2024, p. 8). Despite such recent checklists, changes to reporting practices will be slow, particularly in interdisciplinary fields like MBOR prediction, which draws contributions from authors in cultural economics, information systems, computer science, and other disciplines. However, with respect to the abstract, authors do not need to wait for ML-specific or even MBOR-specific guidelines. For instance, as far back as 2009, the sixth edition of the Publication Manual of the American Psychological Association already mandated that "an abstract should describe […] the basic findings, including effect sizes and confidence intervals and/or statistical confidence levels" (American Psychological Association 2010, p. 26). These statistical requirements should be adapted to the reporting of ML prediction models, necessitating the inclusion of performance metrics in the abstract.

Limitations

The following limitations of our review must be noted. First, regarding the inappropriateness of using strong and leading words, identifying overinterpretation was not objectively possible but required assessment by coding. Based on the critical role of the abstract as a self-contained summary, we coded any qualitative claim lacking support by explicit quantitative results within the abstract as spin. Therefore, our findings regarding the use of strong and leading words should be interpreted as the maximum likely prevalence of this problem. Second, our identification was limited to the abstracts. Previous reviews of other literature have shown that overinterpretation is also relevant in the main texts. Specifically, our analysis of inappropriate extrapolation focused on the abstracts, whereas this reporting practice is also possible in the discussion and conclusion sections. Third, our review is descriptive, as we did not examine how study-level characteristics (e.g., specific modeling approaches, dataset size, or domain) are associated with reporting practices, which could provide hints regarding influencing factors or drivers.

Conclusion

This review contributes to the literature by uncovering prevalent misleading reporting practices in the abstracts of ML studies on movie box office revenue prediction that lead to the overinterpretation of evaluation results. Our findings, particularly the high rate of omission of performance metrics and the inappropriate use of strong and leading language, highlight systemic issues in scholarly communication within this interdisciplinary field. A particular focus is needed on ensuring three critical areas: reporting of interpretable performance metrics in abstracts, using descriptive language appropriately to characterize model performance, and ensuring consistency between abstracts and main texts regarding evaluation metrics. We suggest specific recommendations for the authors, reviewers, and editors to improve the completeness, clarity, and correctness of reporting, which is essential to facilitate the accurate interpretation and reliable integration of evidence from MBOR prediction studies.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1057/s41272-026-00574-9.

Author contributions J.L. and V.S. contributed to the conceptualization of the study. J.L. developed the methodology, administered the project, and wrote the original draft. J.L. and Z.L. conducted the investigation. V.S. contributed to editing the manuscript. All authors reviewed the manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL. No funding was received for conducting this study.

Data availability Data are provided at https://doi.org/10.6084/m9.figshare.30931817.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Ahmad, Ibrahim, Azuraliza Abu Said, Mohd Ridzwan Bakar, and Yaakub, Mohammad Darwich. 2020a. Sequel movie revenue prediction model based on sentiment analysis. *Data Technologies and Applications* 54(5):665–683. https://doi.org/10.1108/DTA-10-2019-0180

Ahmad, Ibrahim, Azuraliza Abu Said, Bakar, and Yaakub Mohd Ridzwan. 2020b. Movie revenue prediction based on purchase intention mining using YouTube trailer reviews. *Information Processing & Management* 57(5):102278. https://doi.org/10.1016/j.ipm.2020.102278

Ahmad, Ibrahim, Azuraliza Abu Said, Mohd Ridzwan Bakar, Yaakub, and Muhammad Shamsuddeen Hassan. 2020c. A survey on machine learning techniques in movie revenue prediction. *SN Computer Science* 1(4). https://doi.org/10.1007/s42979-020-00249-1

American Psychological Association. 2010. *Publication manual of the American psychological association*. 6th ed. Washington, DC: American Psychological Association.

Andaur Navarro, Constanza L., A. A. Johanna, Toshihiko Damen, and Takada et al. 2023. Systematic review finds spin practices and poor reporting standards in studies on machine Learning-Based prediction models. *Journal of Clinical Epidemiology* 158:99–110. https://doi.org/10.1016/j.jclinepi.2023.03.024

Bogaert, Matthias, Michel Ballings, Dirk van den Poel, and Asil Oztekin. 2021. Box office sales and social media: A Cross-Platform comparison of predictive ability and mechanisms. *Decision Support Systems* 147:113517. https://doi.org/10.1016/j.dss.2021.113517

Boutron, Isabelle. 2018. and Philippe Ravaud. Misrepresentation and Distortion of Research in Biomedical Literature. *Proceedings of the National Academy of Sciences* 115 (11): 2613–19. https://doi.org/10.1073/pnas.1710755115

Cals, Jochen W. L., Daniel Kotz. 2013. Effective writing and publishing scientific Papers, part II: Title and abstract. *Journal of Clinical Epidemiology* 66(6):585. https://doi.org/10.1016/j.jclinepi.2013.01.005

Chicco, Davide, and Matthijs J. Warrens, Giuseppe Jurman. 2021. The coefficient of determination R-Squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation. *PeerJ Computer Science* 7:e623. https://doi.org/10.7717/peerj-cs.623

Collins, Gary S., G. M. Karel, Paula Moons, and Dhiman et al. 2024. TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *Bmj* 385:e078378. https://doi.org/10.1136/bmj-2023-078378

Dhiman, Paula, Jie Ma, and Constanza L Andaur Navarro et al. 2023. Overinterpretation of findings in machine learning prediction model studies in oncology: A systematic review. *Journal of Clinical Epidemiology* 157:120–133. https://doi.org/10.1016/j.jclinepi.2023.03.012

Eliashberg, Jehoshua, Anita Elberse, A. A. M. Mark, and Leenders. 2006. The motion picture industry: Critical issues in Practice, current research, and new research directions. *Marketing Science* 25(6):638–661. https://doi.org/10.1287/mksc.1050.0177

Gautam, Nitin, and Shriguru Nayak, Sergey Shebalov. 2021. Machine learning approach to market behavior Estimation with applications in revenue management. *Journal of Revenue and Pricing Management* 20(3):344–350. https://doi.org/10.1057/s41272-021-00317-y

Kempf, Emmanuelle, and Jennifer A. de Beyer, Jonathan Cook et al. 2018. Overinterpretation and misreporting of prognostic factor studies in oncology: A systematic review. *British Journal of Cancer* 119(10):1288–1296. https://doi.org/10.1038/s41416-018-0305-5

Kühl, Niklas, Robin Hirt, Lucas Baier, Björn Schmitz, and Gerhard Satzger. 2021. How to conduct rigorous supervised machine learning in information systems research: The supervised machine learning report card. *Communications of the Association for Information Systems* 48(1):589–615. https://doi.org/10.17705/1CAIS.04845

Lazarus, Clément, Romana Haneef, and Philippe Ravaud, Isabelle Boutron. 2015. Classification and prevalence of spin in abstracts of Non-Randomized studies evaluating an intervention. *BMC Medical Research Methodology* 15:85. https://doi.org/10.1186/s12874-015-0079-x

Lehrer, Steven F., Tian Xie. 2022. The bigger picture: Combining econometrics with analytics improves forecasts of movie success. *Management Science* 68(1):189–210. https://doi.org/10.1287/mnsc.2020.3911

Leukel, Joerg, Luca Scheurer, and Tobias Zimpel. 2025. Overinterpretation of evaluation results in machine learning studies for maize yield prediction: A systematic review. *Computers and Electronics in Agriculture* 230:109892. https://doi.org/10.1016/j.compag.2024.109892

Li, Dawei, and Liu Zhi-Ping. 2022. Predicting Box-Office markets with machine learning methods. *Entropy* 24(5). https://doi.org/10.3390/e24050711

Lu, Yafeng, and Feng Wang, Ross Maciejewski. 2014. Business intelligence from social media: A study from the VAST box office challenge. *IEEE Computer Graphics and Applications* 34(5):58–69. https://doi.org/10.1109/MCG.2014.61

Madongo, Canaan, Zhongjun Tinotenda, and Tang. 2023. and Jahanzeb Hassan. Box-Office Revenue Prediction by Mining Deep Features from Movie Posters and Reviews Using Transformers.

In *2023 6th International Conference on Artificial Intelligence and Pattern Recognition (AIPR 2023)*, 1406–12. New York, NY, USA: ACM.

Mbunge, Elliot, and Stephen Gbenga Fashoto, Happyson Bimha. 2022. Prediction of Box-Office success: A review of trends and machine learning computational models. *International Journal of Business Intelligence and Data Mining* 20(2):192. https://doi.org/10.1504/IJBIDM.2022.120825

McKenzie, Jordi. 2023. The economics of movies (Revisited): A survey of recent literature. *Journal of Economic Surveys* 37(2):480–525. https://doi.org/10.1111/joes.12498

Moriasi, D. N., J. G. Arnold, M. W. van Liew, R. L. Bingner, R. D. Harmel, and T. L. Veith. 2007. Model evaluation guidelines for systematic quantification of accuracy in watershed simulations. *Transactions of the ASABE* 50(3):885–900. https://doi.org/10.13031/2013.23153

Oh, Chong, Yaman Roumani, Joseph K. Nwankpa, and Hu Han-Fen. 2017. Beyond likes and tweets: Consumer engagement behavior and movie box office in social media. *Information & Management* 54(1):25–37. https://doi.org/10.1016/j.im.2016.03.004

Page, Matthew J., E. Joanne, M. McKenzie, Patrick, and Bossuyt et al. 2021. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *Bmj* 372:n71. https://doi.org/10.1136/bmj.n71

Phumchusri, Naragain, Nichakan Phupaichitkun. 2024. Sales prediction hybrid models for retails using promotional pricing strategy as a key demand driver. *Journal of Revenue and Pricing Management* 23(5):461–480. https://doi.org/10.1057/s41272-024-00477-7

Ravid, S., and Abraham. 1999. Information, Blockbusters, and stars: A study of the film industry. *The Journal of Business* 72(4):463–492. https://doi.org/10.1086/209624

Ruan, Dong-ru, Tao Liu, and Kai Gao. 2015. Modelling on Movie Box-Office Prediction Based on LFM Algorithm. In *2015 7th International Conference on Modelling, Identification and Control (ICMIC)*, 1–5: IEEE.

Sheiner, Lewis B., Stuart L. Beal. 1981. Some suggestions for measuring predictive performance. *Journal of Pharmacokinetics and Biopharmaceutics* 9(4):503–512. https://doi.org/10.1007/BF01060893

Shiely, Frances, Kerrie Gallagher, and Seán R. Millar. 2024. How, and Why, science and health researchers read scientific (IMRAD) papers. *PloS One* 19(1):e0297034. https://doi.org/10.1371/journal.pone.0297034

Singh, Vivek, Prashasti Kumar, Mousumi Singh, Jacqueline Karmakar, and Leta, Philipp Mayr. 2021. The journal coverage of web of Science, scopus and dimensions: A comparative analysis. *Scientometrics* 126(6):5113–5142. https://doi.org/10.1007/s11192-021-03948-5

Thelwall, Mike, Pardeep Sud. 2022. Scopus 1900–2020: Growth in Articles, Abstracts, Countries, Fields, and journals. *Quantitative Science Studies* 3(1):37–50. https://doi.org/10.1162/qss_a_00177

Vany, Arthur de. 2003. *Hollywood Economics*: Routledge.

Vasey, Baptiste, Myura Nagendran, and Bruce Campbell et al. 2022. Reporting guideline for the early stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *Bmj* 377:e070904. https://doi.org/10.1136/bmj-2022-070904

Walsh, Ian, Dmytro Fishman, and Dario Garcia-Gasulla et al. 2021. DOME: Recommendations for supervised machine learning validation in biology. *Nature Methods* 18(10):1122–1127. https://doi.org/10.1038/s41592-021-01205-4

Xiao, J., S.-n. Yang, X. Li, H. Jin, and S. Chen. 2016. Quantifying domestic movie revenues using online resources in China. *International Journal of Simulation: Systems Science & Technology*. https://doi.org/10.5013/IJSSST.a.17.02.04

Yavchitz, Amélie, Philippe Ravaud, and Douglas G Altman et al. 2016. A new classification of spin in systematic reviews and Meta-Analyses was developed and ranked according to the severity. *Journal of Clinical Epidemiology* 75:56–65. https://doi.org/10.1016/j.jclinepi.2016.01.020

Zaidi, Alina, Siddhaling Urolagin. 2021. Worldwide gross revenue prediction for Bollywood movies using a hybrid ensemble model. *International Journal of Business Intelligence and Data Mining* 19(1):52. https://doi.org/10.1504/IJBIDM.2021.115952

Zhu, Jun-Jie, and Meiqi Yang, Zhiyong Jason Ren. 2023. Machine learning in environmental research: Common pitfalls and best practices. *Environmental Science & Technology* 57(46):17671–17689. https://doi.org/10.1021/acs.est.3c00026